Orange Condition Classification Report

Classify orange images into the source dataset labels **blackspot**, **canker**, **fresh**, and **grenning**. This is an orange-condition task, not avocado ripeness classification.

Dataset understanding and preparation

The original train/test folders contain 1,090 images. The 991 supplied training images were split with stratification into 842 training and 149 validation samples. The untouched supplied test set contains 99 images.

Class	Train	Test	Total
blackspot	184	22	206
canker	179	22	201
fresh	281	33	314
grenning	347	22	369
Total	991	99	1090

EDA found no unreadable images. blackspot, canker, and grenning images are 800x800 JPGs, while fresh images are variable-size PNGs. This format/resolution difference may act as a dataset shortcut and limits generalization claims.

Methods

The classical baseline combines normalized HSV and CIELAB color histograms with Local Binary Pattern texture features and a class-balanced RBF SVM. The second model uses an ImageNet-pretrained MobileNetV2 with a replacement classification head and frozen feature backbone. Training augmentation and validation-loss early stopping were used. MobileNetV2 was trained for 12 epochs, reached 100% validation accuracy at epoch 10, and had 99.3% validation accuracy at epoch 12. Its minimum validation loss was approximately 0.0904 at epoch 12.

Held-out test results

Both models were evaluated on the same 99 supplied test images. The classical model classified 96/99 correctly; MobileNetV2 classified 94/99 correctly.

Model	Accuracy	Macro precision	Macro recall	Macro F1
Classical SVM	96.97%	97.00%	96.59%	96.57%
MobileNetV2	94.95%	94.61%	94.32%	94.45%

The SVM's three errors were canker images predicted as blackspot. MobileNetV2 confused two blackspot images and two canker images with the other diseased class, and predicted one canker as fresh. Both models classified all grenning images correctly. The SVM's scores were higher on this split by 2.02 percentage points in accuracy and 2.13 percentage points in macro F1; this small test set does not establish performance on a new domain.

Production performance

No production evaluation has been performed. On this CPU, MobileNetV2 inference for a preprocessed 224x224 tensor (batch size 1) measured mean 20.17 ms, median 20.35 ms, and p95 30.34 ms over 30 timed runs after five warmups. Image decoding and resizing are excluded; this is not end-to-end production latency. Evaluate on the intended hardware, camera, lighting, and external data before making production claims.

Drawbacks and future improvements

The test split is small (22-33 images per class), class counts are imbalanced, and image format/resolution correlate with the fresh label. Camera, lighting, background, and framing can cause dataset shift; image appearance also cannot measure internal fruit quality. With more time, collect a larger balanced dataset across sources, verify labels, evaluate on an external test set, compare validation-selected fine-tuning/augmentation strategies, and benchmark the full pipeline on target hardware.

Time spent

No reliable time log was kept for dataset inspection, model experiments, and evaluation, so a measured duration cannot be reported. If required, enter your own truthful estimate from your work history rather than inventing a measurement.

Submission status

The four notebooks have executed successfully; trained models, evaluation outputs, and this report are present. Publishing the repository/notebooks with instructor-approved permissions and recording the requested walkthrough video still require the student's own accounts and actions. Confirm the instructor's rubric and required export format before submission.